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Inventor(s)

Lee; Joanna et al.

LINE DEFECT DETECTION

Abstract

A computing system generates a training data set for training a machine learning model to detect line defects on a surface of a specimen. The computing system trains the machine learning model to detect line defects based on the training data set. The machine learning model is trained to detect line defects by detecting the individual defects represented by the plurality of annotations extending between the starting point and the end point. The computing system determines that the machine learning model has exceeded a threshold level of accuracy. Based on the determining, the computing system deploys the machine learning model to detect line defects on future specimens.

Inventors: Lee; Joanna (Brooklyn, NY), Keith; Jacob (Brooklyn, NY), Doshi; Anuj (Long Island City, NY)

Applicant: Nanotronics Imaging, Inc. (Cuyahoga Falls, OH)

Family ID: 1000007781957

Assignee: Nanotronics Imaging, Inc. (Cuyahoga Falls, OH)

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Background/Summary

TECHNICAL FIELD

[0001] Embodiments disclosed herein generally relate to systems and methods for detecting line defects on a specimen.

BACKGROUND

[0002] Inspection of specimens, such as ceramics, metals, glass, or electronics, for defects and other characteristics is important for manufacturing processes. A defect, such as a line defect, may often be formed on the surface of a specimen. The line defect may be formed during the manufacturing process and is important to detect. Some of these defects are generally longer than other types of defects and may be seen in a single field of view of an imaging system or may span multiple fields of views. In some embodiments, the defect is continuous, i.e., connected, and may be straight or contain curves and/or angles. More generally, this type of defect may be thought of as a defect that is much longer in one dimension than in any other dimensions. The defects as described herein will be referred to as line defects.

SUMMARY

[0003] In some embodiments, a method is disclosed herein. A computing system generates a training data set for training a machine learning model to detect line defects on a surface of a specimen. The generating includes causing an annotation interface to be displayed to a user via a user device of the user, the annotation interface configured for receiving annotations indicating a presence of a defect, causing presentation of a plurality of training images via the annotation interface on the user device, wherein each training image of the plurality of training images comprises a line defect, and receiving annotated images from the user device via the annotation interface, wherein each line defect is annotated using a plurality of annotations that extend between a starting point of the line defect to an end point of the line defect, wherein the annotated images form the training data set. The computing system trains the machine learning model to detect line defects based on the training data set. The machine learning model is trained to detect line defects by detecting the individual defects represented by the plurality of annotations extending between the starting point and the end point. The computing system determines that the machine learning model has exceeded a threshold level of accuracy. Based on the determining, the computing system deploys the machine learning model to detect line defects on future specimens.

[0004] In some embodiments, a system is disclosed herein. The system includes a processor and a memory. The memory has programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations. The operations include generating a training data set for training a machine learning model to detect line defects on a surface of a specimen. The generating includes causing an annotation interface to be displayed to a user via a user device of the user, the annotation interface configured for receiving annotations indicating a presence of a defect, causing presentation of a plurality of training images via the annotation interface on the user device, wherein each training image of the plurality of training images comprises a line defect, and receiving annotated images from the user device via the annotation interface, wherein each line defect is annotated using a plurality of annotations that extend between a starting point of the line defect to an end point of the line defect, wherein the annotated images form the training data set. The operations further include training the machine learning model to detect line defects based on the training data set. The machine learning model is trained to detect line defects by detecting the individual defects represented by the plurality of annotations extending between the starting point and the end point. The operations further include determining that the machine learning model has exceeded a threshold level of accuracy. The operations further include, based on the determining, deploying the machine learning model to detect line defects on future specimens.

[0005] In some embodiments, a non-transitory computer readable medium is disclosed herein. The non-transitory computer readable medium has one or more sequences of instructions stored thereon, which, when executed by a processor, causes a computing system to perform operations.

The operations include generating, by the computing system, a training data set for training a machine learning model to detect line defects on a surface of a specimen. The generating includes causing an annotation interface to be displayed to a user via a user device of the user, the annotation interface configured for receiving annotations indicating a presence of a defect, causing presentation of a plurality of training images via the annotation interface on the user device, wherein each training image of the plurality of training images comprises a line defect, and receiving annotated images from the user device via the annotation interface, wherein each line defect is annotated using a plurality of annotations that extend between a starting point of the line defect to an end point of the line defect, wherein the annotated images form the training data set. The operations further include training, by the computing system, the machine learning model to detect line defects based on the training data set. The machine learning model is trained to detect line defects by detecting the individual defects represented by the plurality of annotations extending between the starting point and the end point. The operations further include determining, by the computing system, that the machine learning model has exceeded a threshold level of accuracy. The operations further include, based on the determining, deploying, by the computing system, the machine learning model to detect line defects on future specimens.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0006] The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the relevant art(s) to make and use embodiments described herein.

[0007] FIG. 1A illustrates a cross-section of a specimen with a line defect, according to example embodiments of the present disclosure.

[0008] FIG. 1B illustrates a top view of a specimen with a line defect, according to example embodiments of the present disclosure.

[0009] FIG. 2 is a block diagram illustrating an exemplary computing environment, according to example embodiments of the present disclosure.

[0010] FIG. 3A illustrates an image captured of a specimen with a line defect labeled, according to example embodiments of the present disclosure.

[0011] FIG. 3B illustrates a line defect labeled in its entirety, according to example embodiments of the present disclosure.

[0012] FIG. 4 is a flowchart for a method for detecting line defects on a specimen using a trained machine learning model, according to example embodiments of the present disclosure.

[0013] FIG. 5A is an exemplary screenshot illustrating an annotated line defect, according to example embodiments.

[0014] FIG. 5B is an exemplary screenshot illustrating a zoomed out view of the annotated line defect of FIG. 5A, according to example embodiments.

[0015] FIG. 6A is a block diagram illustrating a computing device, according to example embodiments of the present disclosure.

[0016] FIG. 6B is a block diagram illustrating a computing device, according to example embodiments of the present disclosure.

[0017] The features of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference

number first appears. Unless otherwise indicated, the drawings provided throughout the disclosure should not be interpreted as to-scale drawings.

DETAILED DESCRIPTION

[0018] The present disclosure is directed to relate to systems and methods for detecting line defects on a specimen using a trained machine learning model.

[0019] Typically, an image taken of the specimen may be annotated for training models to detect such defects. The annotated images, which may be referred to as tagged data, is an important source of information for image classifiers or object detectors, such as deep learning-based models in artificial intelligence (AI). U.S. Pat. No. 11,727,672, which is incorporated by reference, describes generating training sets for training a machine learning model to detect defects on a specimen.

[0020] As those skilled in the art understand, in order to utilize machine learning or artificial intelligence to detect defects on a surface of a specimen, the machine learning or artificial intelligence model needs to be trained to detect such defects. Because line defects are typically longer in length than other defects and may span an entire length of a field of view of an imaging device when under inspection, traditional image annotation techniques for generating a labeled data set to train the machine learning or artificial intelligence model to detect line defects are not as effective. For example, if a line starts to extend beyond a threshold percentage of the field of view (e.g., 33%), the conventional method of annotating line defects becomes less useful, especially if the orientation of the line is angled relative to the angle of the image's pixel grid.

[0021] Conventional annotation techniques for labeling line defects include image segmentation. Image segmentation typically requires a user to assign a label to every pixel in an image of a specimen. As those skilled in the art understand, this process is often time consuming and strenuous as the user needs to manually label every individual pixel in the image. Additionally, using conventional annotation techniques to draw a bounding box or shape around the line defect is often futile because the line defect typically spans an entire field of view of the camera or imaging device capturing an image of the specimen.

[0022] One or more techniques described herein provide an improved approach for training a machine learning model to detect and/or classify line defects on specimens. For example, one or more techniques described herein provide an improved method for labeling line defects on images of specimens to generate a robust training data set for training a machine learning model to detect these defects.

[0023] FIG. 1A illustrates a cross-section of a specimen **100** with a line defect **102**, according to example embodiments. FIG. 1B illustrates a top view of a specimen **100** with a line defect **102**, according to example embodiments.

[0024] A line defect **102** may be formed on the surface of a specimen **100**. In some embodiments, the line defect **102** is a groove. The shape of groove may be a V-shape or a U-shape. The line defect **102** may be formed from one or more lithography and etching operations, mechanical scratching by a diamond tip, by laser etching, or the like. In some embodiments, a depth of the line defect may be in a range from about 20 nm to about 1000 nm. In some embodiments, a depth of the line defect may be in a range from about 100 nm to about 500 nm. In some embodiments, the line defect **102** may span the entire field of view. Accordingly, a given line defect **102** may have parameters associated therewith, such as a length of the line defect, a depth of the line defect, a shape of the line defect, etc.

[0025] FIG. 2 is a block diagram illustrating an exemplary computing environment **200**, according to example embodiments.

[0026] Computing environment **200** may include at least one or more user devices **202** and a back-end computing system **204** communicating via network **205**.

[0027] Network **205** may be of any suitable type, including individual connections via the Internet, such as cellular or Wi-Fi networks. In some embodiments, network **205** may connect terminals,

services, and mobile devices using direct connections, such as radio frequency identification (RFID), near-field communication (NFC), Bluetooth™, low-energy Bluetooth™ (BLE), Wi-Fi™, ZigBee™, ambient backscatter communication (ABC) protocols, USB, WAN, or LAN. Because the information transmitted may be personal or confidential, security concerns may dictate one or more of these types of connection be encrypted or otherwise secured. In some embodiments, however, the information being transmitted may be less personal, and therefore, the network connections may be selected for convenience over security.

[0028] Network **205** may include any type of computer networking arrangement used to exchange data. For example, network **205** may be the Internet, a private data network, virtual private network using a public network and/or other suitable connection(s) that enables components in computing environment **200** to send and receive information between the components of computing environment **200**.

[0029] User device **202** may be operated by a user. In some embodiments, user device **202** may be operated by a user involved in the generation of training data for a machine learning model. User device **202** may be representative of a mobile device, a tablet, a desktop computer, or any computing system having the capabilities described herein. For example, user device **202** may be representative of a mobile device, a tablet, a desktop computer, or any computer system that allows the user to draw shapes and figures on the image, for example, by using an input device, such as, but not limited to, a keyboard, a finger, a mouse, or a stylus.

[0030] User device **202** may include at least application **212**. Application **212** may be representative of an application associated with back-end computing system **204**. In some embodiments, application **212** may be a standalone application associated with back-end computing system **204**. In some embodiments, application **212** may be representative of a web browser configured to communicate with back-end computing system **204**. In some embodiments, user device **202** may communicate over network **205** to request a webpage, for example, from web client application server **214** of back-end computing system **204**. For example, user device **202** may be configured to execute application **212** to label training data for a machine learning model. For example, via application **212**, a user may utilize touchscreen capability of user device **202** to annotate images for training a machine learning model.

[0031] The content that is displayed to user device **202** may be transmitted from web client application server **214** to user device **202**, and subsequently processed by application **212** for display through a graphical user interface (GUI) of user device **202**.

[0032] Back-end computing system **204** may include web client application server **214**, labeling module **216**, and training module **218**. Each of data labeling module **216** and training module **218** may be comprised of one or more software modules. The one or more software modules are collections of code or instructions stored on a media (e.g., memory of back-end computing system **204**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of back-end computing system **204** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that are interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather than as a result of the instructions.

[0033] As stated above, application **212** may allow users to manually annotate an image for the purpose of generating training data to train a machine learning model. Labeling module **216** may be configured to provide user device **202** with a plurality of images of specimens for defect labeling. Generally, the plurality of images of specimens for defect labeling may include at least a line defect present thereon. In some embodiments, the plurality of images of specimens may include other defects present thereon.

[0034] Via application **212**, a user of user device **202** may manually annotate the plurality of

images provided by labeling module **216** to identify defects that are present in the image of the specimen. As indicated above, the labeling of a line defect is typically difficult, due to the length of line defects. Accordingly, to improve upon the labeling process, a user of user device **202** may label a line defect using a plurality of annotations along a length of the line defect. For example, across the length of a line defect, a user may draw a plurality of annotations. In this manner, when the labeled image is used to train machine learning model **226**, the annotations may trick the machine learning model **226** to detect a line defect by detecting a plurality of individual defects across a length of the line defect.

[0035] In some embodiments, the plurality of annotations may overlap. For example, for n-annotations across a length of a line defect, annotation **1** may overlap with annotation **2**, annotation **2** may overlap with annotation **3**, and so on. In some embodiments, the plurality of annotations may not overlap.

[0036] FIG. 5A and FIG. 5B are exemplary screenshots **500**, **550** illustrating an annotated line defect, according to example embodiments. As shown in FIG. 5A, screenshot **500** includes an annotation area **502**. Via annotation area **502**, an end user may annotate an image of a specimen to signal the existence of line defects present thereon. As shown, annotation area **502** includes two groupings of annotations signaling a first line defect **504** and a second line defect **506**.

[0037] FIG. 5B includes an annotation area **552**. The specimen shown in annotation area **552** is a zoomed out version of the specimen, with first line defect **504** and second line defect **506**. As shown, although it appears in FIG. 5A that there are two long lines that have been annotated, when the image is zoomed out, it becomes clear that the two line defects **504**, **506** have been annotated using a plurality of overlapping annotations.

[0038] Referring back to FIG. 2, in some embodiments, labeling a line defect may include specifying the class of defect. For example, a target training process may involve training the machine learning model to detect m-classes of objects or defects, such as line defects. For each class, the user may use a different color to denote the class of the defect. Accordingly, the annotations provided by the user for labeling a line defect may be a separate color from the rest of the annotated defects. In this manner, the machine learning model may learn how to classify defects, in addition to detecting the defects.

[0039] In some embodiments, labeling module **216** may adjust the labels generated by user device **202**. For example, labeling module **216** may be configured to adjust the labels to a standardized format. In some embodiments, the standardized format may include one or more of a shape and size of the annotation. For example, some machine learning models may require annotations to be in form of a bounding box. All the information associated with the original annotations (e.g., pre-transformation/translation) may be stored as meta information for the annotations. This allows the images to be used across multiple annotation formats.

[0040] In some embodiments, such as that discussed in U.S. Pat. No. 11,727,672, which is incorporated by reference herein, labeling module **216** may adjust the labels generated by user device **202** in accordance with a tagging profile and/or error profile associated with a user. For example, based on a learned tagging profile and/or error profile, labeling module **216** may convert or adjust the user's annotations into a standardized format prior to training machine learning model.

[0041] Training module **218** may be configured to train machine learning model **226** to detect line defects using the labeled image sets generated by user device **202** via data labeling module **216**. In some embodiments, machine learning model **226** may be representative of a ResNet based machine learning architecture. In some embodiments, machine learning model **226** may be representative of a transformer based architecture. In some embodiments, machine learning model **226** may be representative of other machine learning models, such as, but are not limited to, random forest model, support vector machines, neural networks, deep learning models, Bayesian algorithms, Temporal Convolutional Networks, and the like.

[0042] As output, training module **218** may generate a trained model **220** to be deployed to detect

line defects on future specimens. Exemplary specimens may include, but are not limited to, substrates, photomasks, optical lenses, and lithography plates.

[0043] Those skilled in the art understand that, in some embodiments, rather than having the components associated with application **212** centrally hosted, such as via back-end computing system **204**, labeling module **216**, training module **218**, and/or trained model **220** may exist locally on user device **202**. In other words, rather than rely on communications with web client application server **214** for accessing functionality of data labeling module **216**, training module **218**, and/or trained model **220**, a user may access such functionality locally via user device **202**, without needing to be connected to network **205**.

[0044] FIG. **3A** illustrates an image **310** captured of a specimen with a line defect **102** labeled, according to example embodiments.

[0045] As shown, user device **202** may take the form of a touchscreen device, such as, but not limited to, a tablet, mobile device, laptop, and the like. User device **202** may include an input interface. In some embodiments, an input interface is provided. A user of user device **202** may use an input device, such as their hand (e.g., at least one finger) or a stylus (e.g., smart pencil device), to provide input to user device **202** via the input interface. In this manner, by simply annotating the image, the user may provide input in the form of where the line defect **102** is located on the image **310**.

[0046] In some embodiments, the line defect **102** may be annotated by the user with multiple annotations **320** labeled across it. For example, a light ring assembly (not pictured) may illuminate the specimen **100** to take an image **310**. Although one image **310** is illustrated in FIG. **3A**, more images **310** may be taken using an illumination system. Within the image **310**, the line defect **102** is visible. The user may draw a small annotation **320** around a portion of the line defect **102** on the user device **202**. The user may make multiple annotations **320** across the entire length of the line defect **102**. The user may either copy and paste the annotation **320** to cover the length of the line defect or use a point and click method to make multiple annotations **320**.

[0047] FIG. **3B** illustrates a line defect **102** labeled in its entirety with annotations **320**, according to an embodiment.

[0048] The annotations **320** may extend the entire length of line defect **102**. Here, the line defect **102** is shown dashed for merely illustration purposes. The annotations **320** cover the line defect **102** in its entirety. This may be done for every line defect **102** that is seen in the image **310**. Depending on the length of the line defect **102**, thousands of annotations may be created across the image **310**.

[0049] By creating multiple annotations **320** around the line defect **102**, the user is effectively tricking the machine learning model to detect the line defect by detecting a multiple individual defects along the length of the line defect. In some embodiments, labeling module **216** may allow for auto-completion of labeling a line defect. For example, based on a color the user selected or a setting, the user can define a starting point of a line defect using a first annotation and an end point of the line defect using the second annotation. Based on this input, labeling module **216** may automate the process of drawing a plurality of annotations along the length of the line defect between the first annotation and the second annotation. In some embodiments, a user may assist with this process by drawing a line between the first annotation and the second annotation to indicate where the plurality of annotations should be populated.

[0050] In some embodiments, using this approach, a machine learning model may also be able to determine attributes of the detected line defect. For example, by labeling line defect **102** with multiple annotations **320**, the machine learning model may be trained to determine the length and depth of the line defect.

[0051] Despite applying this technique to a defect such as a line defect **102**, this technique may also be applied to area type defects. An area may be labeled with multiple annotations to get one large label that may canvas the entire area.

[0052] FIG. **4** is a flowchart for a method **400** for detecting line defects on a specimen using a

trained machine learning model, according to example embodiments. Method **400** may begin at step **402**.

[0053] At step **402**, back-end computing system **204** may cause an annotation interface to be displayed to a user via application **212** executing on user device **202**. The annotation interface may allow the user to review images of specimens, identify defects present on the specimens, and label the defects present thereon.

[0054] At step **404**, back-end computing system **204** may cause a plurality of images to be displayed to via user device **202** for annotation. For example, labeling module **216** may provide a set of images (e.g., image **310**) of specimens for the user to annotate. Generally, the set of images may include images of specimens that have a line defect present thereon. In some embodiments, each image may include other defects present thereon in addition to or in lieu of the line defect.

[0055] At step **406**, back-end computing system **204** may receive annotated images from user device **202**. For example, as discussed above, the user may annotate each image in the set of images with annotations that indicate the presence of a defect. The annotated images may include at least one annotation corresponding to a line defect. For example, a labeled image may include a plurality of annotations that span a length of a line defect to signal the presence of a line defect. In some embodiments, the annotated images may be color-coded to indicate a class of defect represented by the annotation. For example, the plurality of annotations that span a length of a line defect may be labeled a first color, while other defects that are not line defects may be labeled using a second color. In some embodiments, user device **202** may be configured to receive annotations via a stylus or a finger.

[0056] At step **408**, back-end computing system **204** may generate a training data set for training a machine learning model to detect line defects. For example, labeling module **216** may compile the annotated set of images received by user device **202**. In some embodiments, labeling module **216** may perform one or more standardization processes on the annotations generated by the user. For example, labeling module **216** may standardize the user's labels to a predefined shape and/or size. In some embodiments, generating the training data set may include labeling module **216** allocating the annotated set of images between a training data set and a test data set.

[0057] At step **410**, back-end computing system **204** may train a machine learning model to detect line defects based on the training data set. For example, using the training data set, training module **218** may train machine learning model **226** to detect line defects. Generally, training module **218** may train machine learning model **226** until machine learning model **226** exhibits a threshold level of accuracy in detecting line defects. In some embodiments, training machine learning model **226** may include iteratively updating weights and/or hyperparameters of machine learning model **226** until the threshold level of accuracy is achieved.

[0058] Although the above process focuses on detecting line defects, those skilled in the art understand that such training process may result in machine learning model **226** learning to detect defects other than line defects. In some embodiments, machine learning model **226** may be representative of a pre-trained machine learning model that was previously trained to identify defects other than line defects. Thus, the above training process may be performed to supplement or expand the knowledge of machine learning model **226** to include the detection and/or classification of line defects.

[0059] At step **412**, back-end computing system **204** may deploy a trained model to detect defects on a surface of a specimen. For example, once training module **218** determines that machine learning model **226** has achieved a threshold level of accuracy, training module **218** may output trained model **220**. Trained model **220** may then be deployed or hosted on back-end computing system **204**. For example, back-end computing system **204** may then utilize trained model **220** in real-time, near real-time, or periodically to detect and/or classify defects in an unannotated image of a specimen.

[0060] FIG. **6A** illustrates a system bus architecture of computing system **600**, according to

example embodiments. System **600** may be representative of at least an inspection system or a computing system **204**. One or more components of system **600** may be in electrical communication with each other using a bus **605**. System **600** may include a processing unit (CPU or processor) **610** and a system bus **605** that couples various system components including the system memory **615**, such as read only memory (ROM) **620** and random-access memory (RAM) **625**, to processor **610**.

[0061] System **600** may include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **610**. System **600** may copy data from memory **615** and/or storage device **630** to cache **612** for quick access by processor **610**. In this way, cache **612** may provide a performance boost that avoids processor **610** delays while waiting for data. These and other modules may control or be configured to control processor **610** to perform various actions. Other system memory **615** may be available for use as well. Memory **615** may include multiple different types of memory with different performance characteristics. Processor **610** may include any general-purpose processor and a hardware module or software module, such as service **1 632**, service **2 634**, and service **3 636** stored in storage device **630**, configured to control processor **610** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **610** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

[0062] To enable user interaction with the computing system **600**, an input device **645** may represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **635** may also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems may enable a user to provide multiple types of input to communicate with computing system **600**. Communications interface **640** may generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

[0063] Storage device **630** may be a non-volatile memory and may be a hard disk or other types of computer readable media which may store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **625**, read only memory (ROM) **620**, and hybrids thereof.

[0064] Storage device **630** may include services **632**, **634**, and **636** for controlling the processor **610**. Other hardware or software modules are contemplated. Storage device **630** may be connected to system bus **605**. In one aspect, a hardware module that performs a particular function may include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **610**, bus **605**, output device **635** (e.g., display), and so forth, to carry out the function.

[0065] FIG. **6B** illustrates a computer system **650** having a chipset architecture that may represent at least an inspection system or a computing system **204**. Computer system **650** may be an example of computer hardware, software, and firmware that may be used to implement the disclosed technology. System **650** may include a processor **655**, representative of any number of physically and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform identified computations. Processor **655** may communicate with a chipset **660** that may control input to and output from processor **655**.

[0066] In this example, chipset **660** outputs information to output **665**, such as a display, and may read and write information to storage device **670**, which may include magnetic media, and solid-state media, for example. Chipset **660** may also read data from and write data to storage device **675** (e.g., RAM). A bridge **680** for interfacing with a variety of user interface components **685** may be provided for interfacing with chipset **660**. Such user interface components **685** may include a

keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to system **650** may come from any of a variety of sources, machine generated and/or human generated.

[0067] Chipset **660** may also interface with one or more communication interfaces **690** that may have different physical interfaces. Such communication interfaces may include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein may include receiving ordered datasets over the physical interface or be generated by the machine itself by processor **655** analyzing data stored in storage device **670** or storage device **675**. Further, the machine may receive inputs from a user through user interface components **685** and execute appropriate functions, such as browsing functions by interpreting these inputs using processor **655**.

[0068] It may be appreciated that example systems **600** and **650** may have more than one processor **610** or be part of a group or cluster of computing devices networked together to provide greater processing capability.

[0069] While the foregoing is directed to embodiments described herein, other and further embodiments may be devised without departing from the basic scope thereof. For example, aspects of the present disclosure may be implemented in hardware or software or a combination of hardware and software. One embodiment described herein may be implemented as a program product for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein) and may be contained on a variety of computer-readable storage media. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory (ROM) devices within a computer, such as CD-ROM disks readably by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid state random-access memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the disclosed embodiments, are embodiments of the present disclosure.

[0070] It will be appreciated to those skilled in the art that the preceding examples are exemplary and not limiting. It is intended that all permutations, enhancements, equivalents, and improvements thereto are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It is therefore intended that the following appended claims include all such modifications, permutations, and equivalents as fall within the true spirit and scope of these teachings.

Claims

1. A computer-implemented method comprising: generating, by a computing system, a training data set for training a machine learning model to detect line defects on a surface of a specimen, the generating comprising: causing an annotation interface to be displayed to a user via a user device of the user, the annotation interface configured for receiving annotations indicating a presence of a defect, causing presentation of a plurality of training images via the annotation interface on the user device, wherein each training image of the plurality of training images comprises a line defect, receiving annotated images from the user device via the annotation interface, wherein each line defect is annotated using a plurality of annotations that extend between a starting point of the line defect to an end point of the line defect, wherein the annotated images form the training data set; training, by the computing system, the machine learning model to detect line defects based on the training data set, wherein the machine learning model is trained to detect line defects by detecting individual defects represented by the plurality of annotations extending between the starting point

and the end point; determining, by the computing system, that the machine learning model has exceeded a threshold level of accuracy; and based on the determining, deploying, by the computing system, the machine learning model to detect line defects on future specimens.

2. The computer-implemented method of claim 1, wherein generating the training data set comprises: automatically adjusting the plurality of annotations to a standard shape or standard size.

3. The computer-implemented method of claim 1, further comprises: for a training image, detecting a first annotation corresponding to a first starting point of a first line defect, detecting a second annotation corresponding to a first end point of the first line defect, and automatically populating a plurality of annotations between the first annotation and the second annotation.

4. The computer-implemented method of claim 3, further comprising: detecting a presence of an annotated line between the first annotation and the second annotation; and populating the plurality of annotations between the first annotation and the second annotation along the annotated line.

5. The computer-implemented method of claim 1, wherein each of the plurality of annotations corresponding to the line defects are of a first color.

6. The computer-implemented method of claim 5, wherein the annotated images further comprise additional annotations corresponding to defects that are not line defects, each of the additional annotations formed from a second color.

7. The computer-implemented method of claim 6, wherein training, by the computing system, the machine learning model to detect line defects based on the training data set comprises: training the machine learning model to classify the line defects based on the first color and the second color.

8. A system, comprising: a processor; and a memory having programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations comprising: generating a training data set for training a machine learning model to detect line defects on a surface of a specimen, the generating comprising: causing an annotation interface to be displayed to a user via a user device of the user, the annotation interface configured for receiving annotations indicating a presence of a defect, causing presentation of a plurality of training images via the annotation interface on the user device, wherein each training image of the plurality of training images comprises a line defect, receiving annotated images from the user device via the annotation interface, wherein each line defect is annotated using a plurality of annotations that extend between a starting point of the line defect to an end point of the line defect, wherein the annotated images form the training data set; training the machine learning model to detect line defects based on the training data set, wherein the machine learning model is trained to detect line defects by detecting individual defects represented by the plurality of annotations extending between the starting point and the end point; determining that the machine learning model has exceeded a threshold level of accuracy; and based on the determining, deploying the machine learning model to detect line defects on future specimens.

9. The system of claim 8, wherein generating the training data set comprises: automatically adjusting the plurality of annotations to a standard shape or standard size.

10. The system of claim 8, wherein the operations further comprise: for a training image, detecting a first annotation corresponding to a first starting point of a first line defect, detecting a second annotation corresponding to a second end point of the first line defect, and automatically populating a plurality of annotations between the first annotation and the second annotation.

11. The system of claim 10, wherein the operations further comprise: detecting a presence of an annotated line between the first annotation and the second annotation; and populating the plurality of annotations between the first annotation and the second annotation along the annotated line.

12. The system of claim 8, wherein each of the plurality of annotations corresponding to the line defects are of a first color.

13. The system of claim 12, wherein the annotated images further comprise additional annotations corresponding to defects that are not line defects, each of the additional annotations formed from a second color.

14. The system of claim 13, wherein training the machine learning model to detect line defects based on the training data set comprises: training the machine learning model to classify the line defects based on the first color and the second color.

15. A non-transitory computer readable medium having one or more sequences of instructions stored thereon, which, when executed by a processor, causes a computing system to perform operations comprising: generating, by the computing system, a training data set for training a machine learning model to detect line defects on a surface of a specimen, the generating comprising: causing an annotation interface to be displayed to a user via a user device of the user, the annotation interface configured for receiving annotations indicating a presence of a defect, causing presentation of a plurality of training images via the annotation interface on the user device, wherein each training image of the plurality of training images comprises a line defect, receiving annotated images from the user device via the annotation interface, wherein each line defect is annotated using a plurality of annotations that extend between a starting point of the line defect to an end point of the line defect, wherein the annotated images form the training data set; training, by the computing system, the machine learning model to detect line defects based on the training data set, wherein the machine learning model is trained to detect line defects by detecting individual defects represented by the plurality of annotations extending between the starting point and the end point; determining, by the computing system, that the machine learning model has exceeded a threshold level of accuracy; and based on the determining, deploying, by the computing system, the machine learning model to detect line defects on future specimens.

16. The non-transitory computer readable medium of claim 15, further comprises: for a training image, detecting a first annotation corresponding to a first starting point of a first line defect, detecting a second annotation corresponding to a second end point of the first line defect, and automatically populating a plurality of annotations between the first annotation and the second annotation.

17. The non-transitory computer readable medium of claim 16, further comprising: detecting a presence of an annotated line between the first annotation and the second annotation; and populating the plurality of annotations between the first annotation and the second annotation along the annotated line.

18. The non-transitory computer readable medium of claim 15, wherein each of the plurality of annotations corresponding to the line defects are of a first color.

19. The non-transitory computer readable medium of claim 18, wherein the annotated images further comprise additional annotations corresponding to defects that are not line defects, each of the additional annotations formed from a second color.

20. The non-transitory computer readable medium of claim 19, wherein training, by the computing system, the machine learning model to detect line defects based on the training data set comprises: training the machine learning model to classify the line defects based on the first color and the second color.
